

MAT21-4

Discrete Stochastic Models & Concentration

Albert School — 22 hours

Preface

You arrive at this course already fluent in three languages. From *Discrete Mathematics* (MAT11-4) you know how to slice a set into an **equivalence partition** and reason case by case — “given that we are in block A , what fraction of A has property B ?” From *Probability & Statistics I* (MAT21-2) you know random variables, expectation, variance, the classic discrete laws, and the total-probability formula. From *Linear Algebra I* (MAT11-3) you know how to multiply matrices and raise them to powers, and you have already met finite state transitions as matrices.

This course fuses the three. **Conditional expectation** is nothing but the weighted-average-over-a-partition idea of MAT11-4, now applied to a random variable instead of a count. **Markov chains** are the matrix powers of MAT11-3, now read as probabilities. And the new ingredient — **concentration** — answers the question MAT21-2 raised but never closed: knowing the mean and variance of an estimator, *how many observations* do we actually need before we may trust it? We will keep the apparatus light: no sigma-algebras, no formal filtrations. Those arrive, built from scratch, in S3 and S6. Here every idea is introduced on a finite or countable space where you can, in principle, list every outcome.

Each chapter opens with a concrete example you can compute by hand, then states the general principle, then flags the mistakes students reliably make. Read with a pen.

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1 Conditional expectation on finite partitions

1.1 From “average over a block” to a random variable

Worked example — A warm-up you can already do. Roll a fair die, $\Omega = \{1, \dots, 6\}$, and let $X(\omega) = \omega$ be the face. A friend, without showing you the die, tells you only whether the result is *low* ($\in \{1, 2, 3\}$) or *high* ($\in \{4, 5, 6\}$). What is your best guess for X ?

If told “low”, you average 1, 2, 3 to get 2. If told “high”, you average 4, 5, 6 to get 5. Your guess is therefore a *new random variable*: it equals 2 on the block $\{1, 2, 3\}$ and 5 on the block $\{4, 5, 6\}$. That variable is the conditional expectation $E[X \mid \mathcal{P}]$. It is the heart of this chapter.

We work on a finite probability space (Ω, P) and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. A **partition** of Ω is a family $\mathcal{P} = \{B_1, \dots, B_k\}$ of non-empty disjoint events with $\bigcup_i B_i = \Omega$ and $P(B_i) > 0$. Exactly as in MAT11-4, a partition encodes a piece of information: **which block occurred**.

Definition 1 (Conditional expectation given a block) For an event B with $P(B) > 0$,

$$E[X \mid B] = \sum_x x P(X = x \mid B) = \frac{1}{P(B)} \sum_{\omega \in B} X(\omega) P(\omega).$$

It is the ordinary expectation of X , but re-weighted so that the block B now carries all the probability.

Definition 2 (Conditional expectation given a partition) The conditional expectation of X given $\mathcal{P} = \{B_1, \dots, B_k\}$ is the random variable $E[X \mid \mathcal{P}]$ that is **constant on each block**, taking the value $E[X \mid B_i]$ throughout B_i :

$$E[X \mid \mathcal{P}](\omega) = E[X \mid B_i] \quad \text{if } \omega \in B_i.$$

The key mental shift — and the one students find hardest — is that $E[X \mid \mathcal{P}]$ is *not a number* but a *function of the outcome*. It is the coarsened, “blurred” version of X you would report if you could see only which block occurred.

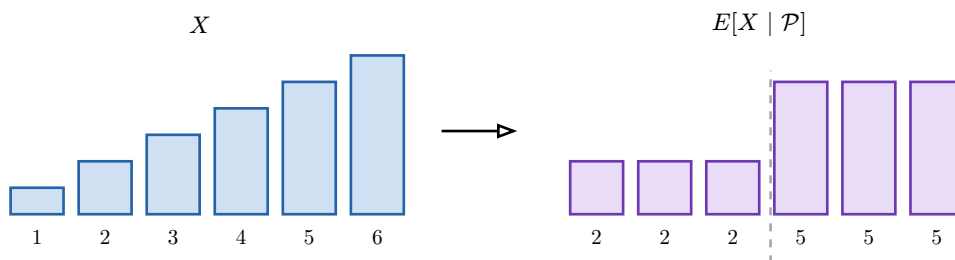


Figure 1: Conditioning on the low/high partition replaces the six distinct values of X (left) by the block averages 2 and 5 (right): a coarser random variable, constant on each block.

Remark The block-averaging always preserves the overall mean: $E[E[X \mid \mathcal{P}]] = E[X]$. In the die example, $\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 5 = 3.5 = E[X]$. We promote this observation to a theorem in the next chapter.

1.2 Partition refinement and the tower property

Computation proceeds by **partition refinement**: to evaluate $E[X \mid \mathcal{P}]$ you restrict to one block at a time and average X over it. Refinement is also the right language for *gaining information in stages*.

Definition 3 (Refinement) A partition $\mathcal{Q}$ **refines** $\mathcal{P}$ (written $\mathcal{Q} \preceq \mathcal{P}$) if every block of $\mathcal{Q}$ is contained in some block of $\mathcal{P}$. Then $\mathcal{Q}$ is *finer* (more informative) and $\mathcal{P}$ is *coarser*.

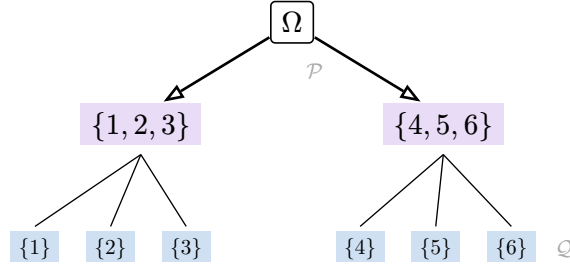


Figure 2: The singleton partition $\mathcal{Q}$ (bottom) refines the low/high partition $\mathcal{P}$ (middle), which in turn refines the trivial partition $\{\Omega\}$ (top). Information increases downward.

Theorem 4 (Tower property) If $\mathcal{Q}$ refines $\mathcal{P}$, then

$$E[E[X \mid \mathcal{Q}] \mid \mathcal{P}] = E[X \mid \mathcal{P}].$$

In particular, taking $\mathcal{P} = \{\Omega\}$,

$$E[E[X \mid \mathcal{Q}]] = E[X].$$

Read it as a *consistency of averaging*: averaging a fine-grained guess back up to a coarse partition returns exactly the coarse guess. You cannot create information by first refining and then re-coarsening.

Worked example — Verifying the tower property on the die. Take $\mathcal{Q} =$ singletons (so $E[X \mid \mathcal{Q}] = X$, since each block pins down ω) and $\mathcal{P} = \{\{1, 2, 3\}, \{4, 5, 6\}\}$. Then $E[E[X \mid \mathcal{Q}] \mid \mathcal{P}] = E[X \mid \mathcal{P}]$, which is 2 on the low block and 5 on the high block — the same variable we built by hand in the warm-up.

Common pitfall Three recurring mistakes.

- **Treating $E[X \mid \mathcal{P}]$ as a number.** It is a random variable; only $E[X \mid B_i]$ for a single block is a number.
- **Inverting the tower.** The inner conditioning must be on the *finer* partition: $E[E[X \mid \mathcal{Q}] \mid \mathcal{P}]$ collapses to $E[X \mid \mathcal{P}]$, but $E[E[X \mid \mathcal{P}] \mid \mathcal{Q}]$ does *not* simplify the same way.
- **Forgetting the weights.** $E[X] = \sum_i E[X \mid B_i] P(B_i)$, not the plain average of the block means — unless the blocks are equiprobable.

2 Variance decomposition: explained vs. residual

2.1 The two questions a partition answers

Conditioning on a partition asks two questions at once. The law of total *expectation* asks: **on average, what is X ?** The law of total *variance* asks: **how much of X 's variability does the partition explain, and how much remains inside the blocks?** This “explained vs. residual” split is the same idea you will later meet as R^2 in regression.

Theorem 5 (Law of total expectation) For a partition $\mathcal{P} = \{B_1, \dots, B_k\}$,

$$E[X] = \sum_{i=1}^k E[X \mid B_i] P(B_i) = E[E[X \mid \mathcal{P}]].$$

Theorem 6 (Law of total variance)

$$\text{Var}(X) = \underbrace{E[\text{Var}(X \mid \mathcal{P})]}_{\text{residual: within-block}} + \underbrace{\text{Var}(E[X \mid \mathcal{P}])}_{\text{explained: between-block}}.$$

Here $\text{Var}(X \mid \mathcal{P})$ is the random variable equal to the within-block variance $\text{Var}(X \mid B_i)$ on B_i ; its expectation is the average spread that survives *after* you learn the block. The second term measures how far the block means scatter around the grand mean — the variability the partition *accounts for*.

Worked example — Two coins — splitting the variance. Pick one of two coins uniformly: coin L lands heads with probability 0.2, coin H with probability 0.8. Flip the chosen coin once and let $X \in \{0, 1\}$ be the result; let $\mathcal{P}$ record which coin was picked.

- **Residual.** Each coin alone has variance $p(1-p) = 0.2 \cdot 0.8 = 0.16$, so $E[\text{Var}(X \mid \mathcal{P})] = 0.16$.
- **Explained.** The block means are 0.2 and 0.8 around the grand mean 0.5, giving $\text{Var}(E[X \mid \mathcal{P}]) = \frac{1}{2}(0.2 - 0.5)^2 + \frac{1}{2}(0.8 - 0.5)^2 = 0.09$.
- **Total.** $\text{Var}(X) = 0.16 + 0.09 = 0.25$, matching $0.5 \cdot 0.5$ directly.

Knowing *which coin* explains $\frac{0.09}{0.25} = 36\%$ of the variability; the rest is the irreducible randomness of a single flip.

Common pitfall The residual term is $E[\text{Var}(X \mid \mathcal{P})]$ — the *expected* within-block variance, averaged over blocks with their probabilities — not the variance of any one block, and not $\text{Var}(E[\text{Var}(X \mid \mathcal{P})])$. Mixing up which term gets the outer expectation and which gets the outer variance is the single most common slip here.

3 Martingale intuition: the fair game

3.1 Betting without an edge

Worked example — A fair coin game. You bet on fair coin flips, winning or losing one euro each round. Let X_n be your fortune after n rounds. Whatever has happened so far, your expected fortune after the next flip is exactly what you hold now: the game has no drift. A

sequence with this “no expected change” property is a **martingale** — the mathematical model of a *fair game*.

Definition 7 (Martingale (fair game)) A sequence $X_0, X_1, X_2, \dots$ is a martingale if for every n ,

$$E[X_{n+1} \mid X_0, \dots, X_n] = X_n.$$

The conditional expected next value equals the current value: on average the game neither gains nor loses. (We condition on the observed history $X_0, \dots, X_n$ informally, exactly as the partition “which path so far” — no filtration formalism is needed at this level.)

Two canonical examples carry us through the rest of the course.

- **Symmetric random walk:** $S_n = \sum_{i=1}^n \xi_i$ with $\xi_i = \pm 1$ equally likely and independent. Since $E[\xi_{n+1}] = 0$ independently of the past, $E[S_{n+1} \mid S_0, \dots, S_n] = S_n + E[\xi_{n+1}] = S_n$.
- **Doob martingale:** fix a target random variable Y and a refining sequence of partitions $\mathcal{P}_0 \not\preceq \dots$; set $X_n = E[Y \mid \mathcal{P}_n]$. The tower property makes (X_n) a martingale: your best estimate of Y updates, on average, to itself as information accrues.

3.2 Optional stopping and where it breaks

A **stopping time** T is a random time you can recognise as it happens: whether “ $T \leq n$ ” holds depends only on $X_0, \dots, X_n$, not on the future. “Stop at the first time my fortune hits 10” is a stopping time; “stop one step before the peak” is not. The fair-game intuition suggests no stopping rule can beat a fair game:

Theorem 8 (Optional stopping (conditions)) If (X_n) is a martingale and T a stopping time, then $E[X_T] = E[X_0]$ provided *at least one* of the following holds:

1. T is bounded: $T \leq N$ for a constant N ; or
2. the increments $|X_{n+1} - X_n|$ are bounded and $E[T] < \infty$; or
3. $X_{n \wedge T}$ is dominated by an integrable random variable.

Common pitfall The conditions are not decoration: drop them all and the conclusion fails. The famous **doubling (“martingale”) strategy** is the trap.

Worked example — A counterexample where optional stopping fails. Take the symmetric random walk S_n (a martingale, $S_0 = 0$) and stop at the first time it reaches +1: $T = \inf\{n : S_n = 1\}$. The walk is recurrent, so $T < \infty$ almost surely and $S_T = 1$. Hence

$$E[S_T] = 1 \neq 0 = E[S_0].$$

There is no contradiction: $E[T] = \infty$, the increments-bounded-and-finite-mean condition fails, and so does every other. The “guaranteed” euro is borrowed against an unbounded — and unbounded-time — debt. This is exactly why no betting system beats a fair casino game in practice.

4 Random walks

4.1 Setup and a sample path

Let $S_0 = k$ and $S_n = k + \sum_{i=1}^n \xi_i$, where the steps ξ_i are independent, $\xi_i = +1$ with probability p and $\xi_i = -1$ with probability $q = 1 - p$. The walk is **symmetric** if $p = \frac{1}{2}$ and **biased** otherwise.

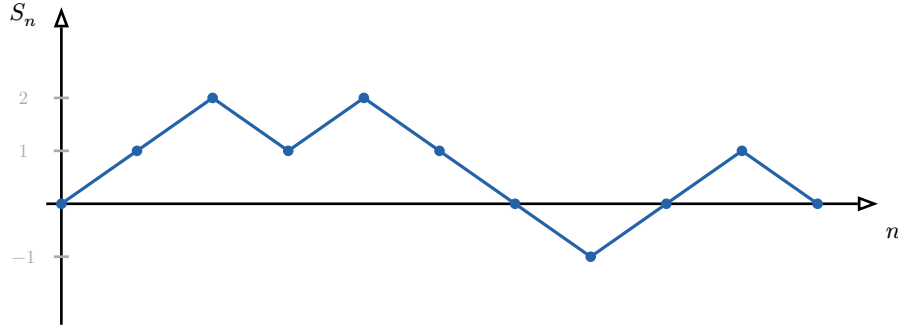


Figure 3: One sample path of a symmetric random walk started at 0, with steps ± 1 . Each upward segment is a $+1$, each downward a -1 .

4.2 Recurrence vs. transience

Proposition 9 (Return behaviour on the integers) The symmetric walk ($p = \frac{1}{2}$) on $\mathbb{Z}$ is **recurrent**: started at 0 it returns to 0 with probability 1. The biased walk ($p \neq \frac{1}{2}$) is **transient**: it returns with probability < 1 and drifts to $+\infty$ (if $p > \frac{1}{2}$) or $-\infty$ (if $p < \frac{1}{2}$).

Remark Recurrence does not mean “returns quickly”. The symmetric walk returns with certainty, yet the *expected* time to return is infinite — the same pathology that broke optional stopping above.

4.3 Gambler’s ruin: absorbing barriers

Worked example — The gambler’s question. You start with k euros, bet one euro per fair round, and play until you either go broke (0) or reach a target N . What is the chance you hit the target first? And how long will you play on average?

The states 0 and N are **absorbing**: once reached, the walk stays there forever. The interior states $1, \dots, N-1$ are transient.

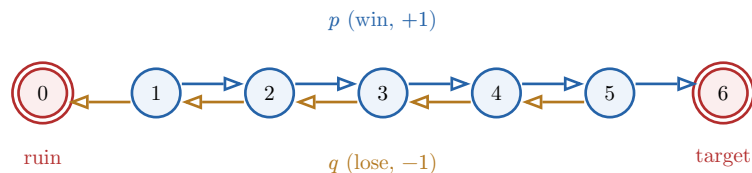


Figure 4: Gambler’s ruin as a walk on $\{0, \dots, N\}$. The barriers 0 and N (double circles) are absorbing; from each interior state the walk steps right with probability p , left with probability q .

Proposition 10 (Ruin / success probabilities) Let $h_k = P(\text{reach } N \text{ before } 0 \mid \text{start at } k)$ and $r = \frac{q}{p}$. Then

$$h_k = \begin{cases} \frac{k}{N} & \text{if } p = \frac{1}{2} \\ \frac{r^k - 1}{r^N - 1} & \text{if } p \neq \frac{1}{2}. \end{cases}$$

Proposition 11 (Expected duration (symmetric case)) Let τ_k be the expected number of steps until absorption, started at k . For $p = \frac{1}{2}$,

$$E[\tau_k] = k(N - k).$$

4.4 First-step analysis: the universal technique

Every quantity above is obtained the same way — **first-step analysis**, the probabilistic cousin of the recurrences you solved in MAT11-4. Condition on the very first step:

$$h_k = p h_{k+1} + q h_{k-1}, \quad h_0 = 0, \quad h_N = 1.$$

This is a linear recurrence in k ; solving it (constant-coefficient, exactly the MAT11-4 method) yields the closed forms above. For the duration, conditioning on the first step costs one move:

$$E[\tau_k] = 1 + p E[\tau_{k+1}] + q E[\tau_{k-1}], \quad E[\tau_0] = E[\tau_N] = 0.$$

Common pitfall When $p = \frac{1}{2}$ the recurrence $h_k = \frac{1}{2}h_{k+1} + \frac{1}{2}h_{k-1}$ has a *repeated* root, so the solution is *linear* ($h_k = \frac{k}{N}$), not geometric. Blindly writing $h_k = \frac{r^k - 1}{r^N - 1}$ with $r = 1$ gives $\frac{0}{0}$. Treat the symmetric case separately.

5 Markov chains

5.1 Memorylessness and the transition matrix

Worked example — Weather as a chain. Suppose tomorrow’s weather depends only on today’s, not on the whole past week: if today is sunny, tomorrow is sunny with probability 0.7 and rainy with 0.3; if rainy, sunny with 0.4 and rainy with 0.6. Collect these into a matrix

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix},$$

rows = “today”, columns = “tomorrow”. You already know from MAT11-3 that P^n gives n -step transitions; now its entries are *probabilities*.

Definition 12 (Markov chain and transition matrix) A sequence (X_n) on a finite or countable state space S is a **Markov chain** if it is *memoryless*:

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j \mid X_n = i) =: P_{ij}.$$

The **transition matrix** $P = (P_{ij})$ has non-negative entries and each row sums to 1 (a *stochastic* matrix). The n -step probabilities are the entries of P^n : $P(X_n = j \mid X_0 = i) = (P^n)_{ij}$.

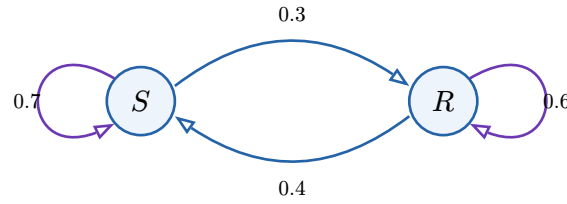


Figure 5: State diagram of the two-state weather chain: nodes are states, arrows carry transition probabilities, and the loops are the “stay” probabilities. Each node’s outgoing probabilities sum to 1.

5.2 Classification of states

- A state i is **absorbing** if $P_{ii} = 1$ (the chain can never leave).
- i is **recurrent** if, started from i , the chain returns to i with probability 1; otherwise it is **transient**.
- The chain is **irreducible** if every state is reachable from every other state (the diagram is strongly connected).
- The **period** of i is $\gcd\{n \geq 1 : (P^n)_{ii} > 0\}$; the chain is **aperiodic** if every state has period 1. A single self-loop $P_{ii} > 0$ forces period 1.

5.3 Stationary distributions

Worked example — The long-run weather. Run the weather chain for a long time: what fraction of days are sunny? Intuition (confirmed below) says the distribution settles to a fixed vector π unchanged by one more day — a row vector with $\pi P = \pi$.

Definition 13 (Stationary distribution) A row vector $\pi = (\pi_i)$ with $\pi_i \geq 0$ and $\sum_i \pi_i = 1$ is a **stationary distribution** if

$$\pi P = \pi.$$

It is a left eigenvector of P with eigenvalue 1 (compare the eigenvector language of MAT11-3), normalised to a probability vector.

Theorem 14 (Convergence to equilibrium) For an **irreducible, aperiodic** chain on a **finite** state space, a stationary distribution π exists, is unique, and is the limit of the row distributions:

$$P(X_n = j) \rightarrow \pi_j \quad \text{as } n \rightarrow \infty,$$

regardless of the starting state.

Worked example — Solving the weather chain. With $P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$, solve $\pi P = \pi$ with $\pi_S + \pi_R = 1$: from $0.7\pi_S + 0.4\pi_R = \pi_S$ we get $0.4\pi_R = 0.3\pi_S$, so $\pi_S : \pi_R = 4 : 3$ and

$$\pi = \left(\frac{4}{7}, \frac{3}{7} \right) \approx (0.571, 0.429).$$

In the long run about 57% of days are sunny, whatever today’s weather.

Common pitfall Irreducibility and aperiodicity are both required. The chain $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is irreducible but has period 2: started at state 1 it is at 1 on even steps and 2 on odd steps, so $P(X_n = 1)$ oscillates and never converges, even though $\pi = (\frac{1}{2}, \frac{1}{2})$ is stationary. Stationary $\neq$ limiting unless the chain is also aperiodic.

6 Concentration inequalities

6.1 Turning “mean and variance” into tail bounds

MAT21-2 gave you the mean and variance of an estimator but stopped short of the practical payoff: a *guarantee* that the estimator is close to the truth. Concentration inequalities supply exactly that — they bound the probability of a large deviation using only crude information about X .

Theorem 15 (Markov's inequality) If $X \geq 0$ and $a > 0$, then

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Proof. Since $X \geq 0$, the indicator inequality $a \mathbf{1}_{X \geq a} \leq X$ holds pointwise (on $X \geq a$ the left side is $a \leq X$; elsewhere it is $0 \leq X$). Taking expectations, $a P(X \geq a) \leq E[X]$; divide by a . $\square$

Theorem 16 (Chebyshev's inequality) If X has mean μ and finite variance σ^2 , then for $k > 0$,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Proof. Apply Markov's inequality to the non-negative variable $(X - \mu)^2$ at threshold k^2 :

$$P(|X - \mu| \geq k) = P((X - \mu)^2 \geq k^2) \leq \frac{E[(X - \mu)^2]}{k^2} = \frac{\sigma^2}{k^2}.$$

$\square$

Remark Chebyshev is *derived* from Markov, not postulated — the standard pattern of this chapter: get a sharper bound by feeding Markov a cleverly chosen non-negative function of X . Hoeffding pushes this to its limit by using the exponential $e^{\lambda X}$.

6.2 Exponential bounds: Chernoff and Hoeffding

For *sums of bounded independent* variables, polynomial decay (Chebyshev, $\frac{1}{k^2}$) is wasteful: the true tails decay *exponentially*. The **Chernoff method** extracts this by applying Markov to $e^{\lambda S}$:

$$P(S \geq a) \leq e^{-\lambda a} E[e^{\lambda S}], \quad \text{then optimise over } \lambda > 0.$$

Carrying it out gives two named results: for $\frac{0}{1}$ summands it yields the **Chernoff bound**, and for general bounded summands it yields **Hoeffding's bound**.

Proposition 17 (Chernoff bound for sums of indicators) Let $S = \sum_{i=1}^n X_i$ with the $X_i \in \{0, 1\}$ independent, and let $\mu = E[S]$. Then for every $\delta > 0$,

$$P(S \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}} \right)^\mu,$$

and the convenient relaxation $P(S \geq (1 + \delta)\mu) \leq \exp\left(-\mu \frac{\delta^2}{3}\right)$ holds for $0 < \delta \leq 1$. The lower tail satisfies $P(S \leq (1 - \delta)\mu) \leq \exp\left(-\mu \frac{\delta^2}{2}\right)$.

Worked example — Applying the Chernoff bound. Toss a fair coin $n = 100$ times; S counts heads, so $\mu = 50$. How unlikely is it to see at least 75 heads? Here $(1 + \delta)\mu = 75$ gives $\delta = \frac{1}{2}$, so

$$P(S \geq 75) \leq \exp\left(-\mu \frac{\delta^2}{3}\right) = \exp\left(-50 \cdot \frac{0.25}{3}\right) \approx 0.0155.$$

The chance of a 75%-heads run is at most about 1.6% — an exponential decay in the number of tosses, far sharper than Chebyshev's $\frac{1}{k^2}$ here.

Theorem 18 (Hoeffding's bound) Let $X_1, \dots, X_n$ be independent with $a_i \leq X_i \leq b_i$, and $S = \sum_{i=1}^n X_i$. Then for every $t > 0$,

$$P(S - E[S] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right),$$

and the same bound holds for the lower tail $P(S - E[S] \leq -t)$.

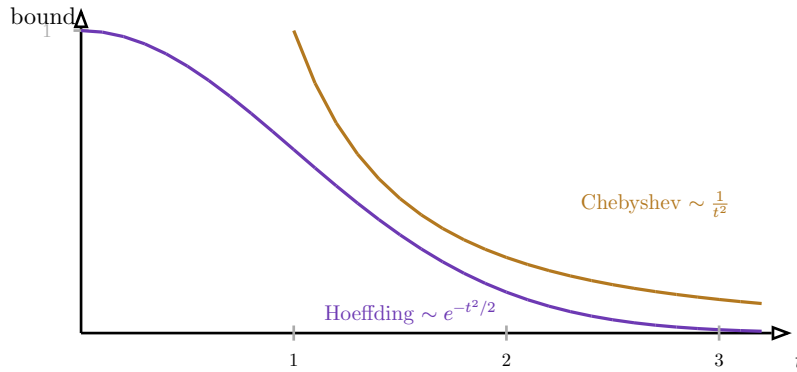


Figure 6: Tail-bound decay against deviation t (in standard-deviation units). Chebyshev's polynomial $\frac{1}{t^2}$ (orange) sits far above Hoeffding's exponential $e^{-t^2/2}$ (purple): both are valid upper bounds, but for sums of bounded variables the exponential bound is dramatically tighter.

Common pitfall Hoeffding needs *independence* and *bounded* summands. With heavy tails or strong dependence the exponential bound can be wildly violated; Chebyshev, needing only a finite variance, still holds. Match the inequality to the hypotheses you actually have.

7 Quantitative uncertainty control

7.1 From a bound to a sample size

The exponential bound pays off as a concrete **sample-size rule**. Let $X_1, \dots, X_n$ be independent with values in $[0, 1]$, common mean μ , and sample mean $\bar{X} = \frac{1}{n} \sum_i X_i$. Applying Hoeffding to $S = \sum_i X_i$ (here $b_i - a_i = 1$) and rescaling by n ,

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2).$$

Proposition 19 (Minimum sample size) To guarantee $P(|\bar{X} - \mu| \geq \varepsilon) \leq \delta$ it suffices to take

$$n \geq \frac{1}{2\varepsilon^2} \ln\left(\frac{2}{\delta}\right).$$

Proof. Set the right-hand bound $2 \exp(-2n\varepsilon^2) \leq \delta$, take logarithms, and solve for n : $2n\varepsilon^2 \geq \ln(\frac{2}{\delta})$, i.e. $n \geq \ln(\frac{2}{\delta})/(2\varepsilon^2)$. $\square$

This is the “confidence $1 - \delta$, tolerance ε ” calculation in its purest form. Notice the two scalings: halving the error ε costs *four times* the data ($\frac{1}{\varepsilon^2}$), while raising the confidence costs only *logarithmically* in $\frac{1}{\delta}$ — cheap confidence, expensive precision.

Worked example — A polling-style computation. For tolerance $\varepsilon = 0.05$ and confidence $1 - \delta = 0.95$ ($\delta = 0.05$),

$$n \geq \frac{1}{2 \cdot 0.05^2} \ln\left(\frac{2}{0.05}\right) = 200 \ln(40) \approx 737.$$

About 740 bounded observations suffice to pin the mean to within 0.05, 19 times out of 20 — *independently of* the underlying distribution.

Remark Hoeffding’s $n \approx 737$ is deliberately conservative: it is distribution-free. A normal-approximation calculation (which MAT21-2 hinted at) would ask for fewer observations but assumes a limit that a finite sample only approximates. Concentration trades sharpness for an honest, non-asymptotic guarantee — the right default when you cannot certify normality.

8 Python mini-labs

These labs close the loop: implement each model, estimate its tail behaviour by simulation, and hold the estimate up against the theory of the preceding chapters. The point is to *see* recurrence, convergence, and exponential concentration happen, and to feel how loose or tight each bound is.

Exercises

1. Simulate a random walk. Generate ± 1 steps and accumulate S_n . Estimate the gambler’s-ruin probability h_k of reaching N before 0 by Monte Carlo and compare to the closed form k/N (symmetric) and $(r^k - 1)/(r^N - 1)$ (biased). Plot a few sample paths like the figure in Chapter 4.

2. Watch a Markov chain converge. Iterate the row distribution $\mu_{n+1} = \mu_n P$ for the weather chain and confirm it approaches $\pi = (\frac{4}{7}, \frac{3}{7})$ from any start. Then switch to $P =$

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and observe that the distribution oscillates and never converges — aperiodicity matters.

3. Empirically verify concentration. For a sum S of n independent bounded variables, estimate $P(S - E[S] \geq t)$ by repeated simulation and overlay the observed tail frequency on Hoeffding's bound $\exp(-2t^2 / \sum (b_i - a_i)^2)$. Check that the simulated tail stays below the bound.

4. Compare the bounds. For a fixed deviation, tabulate the Chebyshev bound, the Hoeffding bound, and the simulated tail probability. Confirm that Hoeffding is exponentially tighter while *both* remain valid upper bounds — the figure in Chapter 7, made empirical.

5. Stress-test optional stopping. Simulate the symmetric walk as a martingale. For a bounded stopping time $T \leq N$, estimate $E[S_T]$ and check $E[S_T] \approx S_0$. Then take the unbounded doubling-strategy stopping time and watch $E[S_T]$ drift away from S_0 — optional stopping failing before your eyes.

6. Sample-size in practice. Pick a target (ε, δ) , compute the Hoeffding sample size n , then simulate to estimate the true coverage. Verify the achieved error exceeds ε at most a δ fraction of the time, and note how much slack the distribution-free bound leaves.